


# Assessment of Direct Economic Cost of Delay at Texas-Mexico Commercial LPOE

February 02, 2023

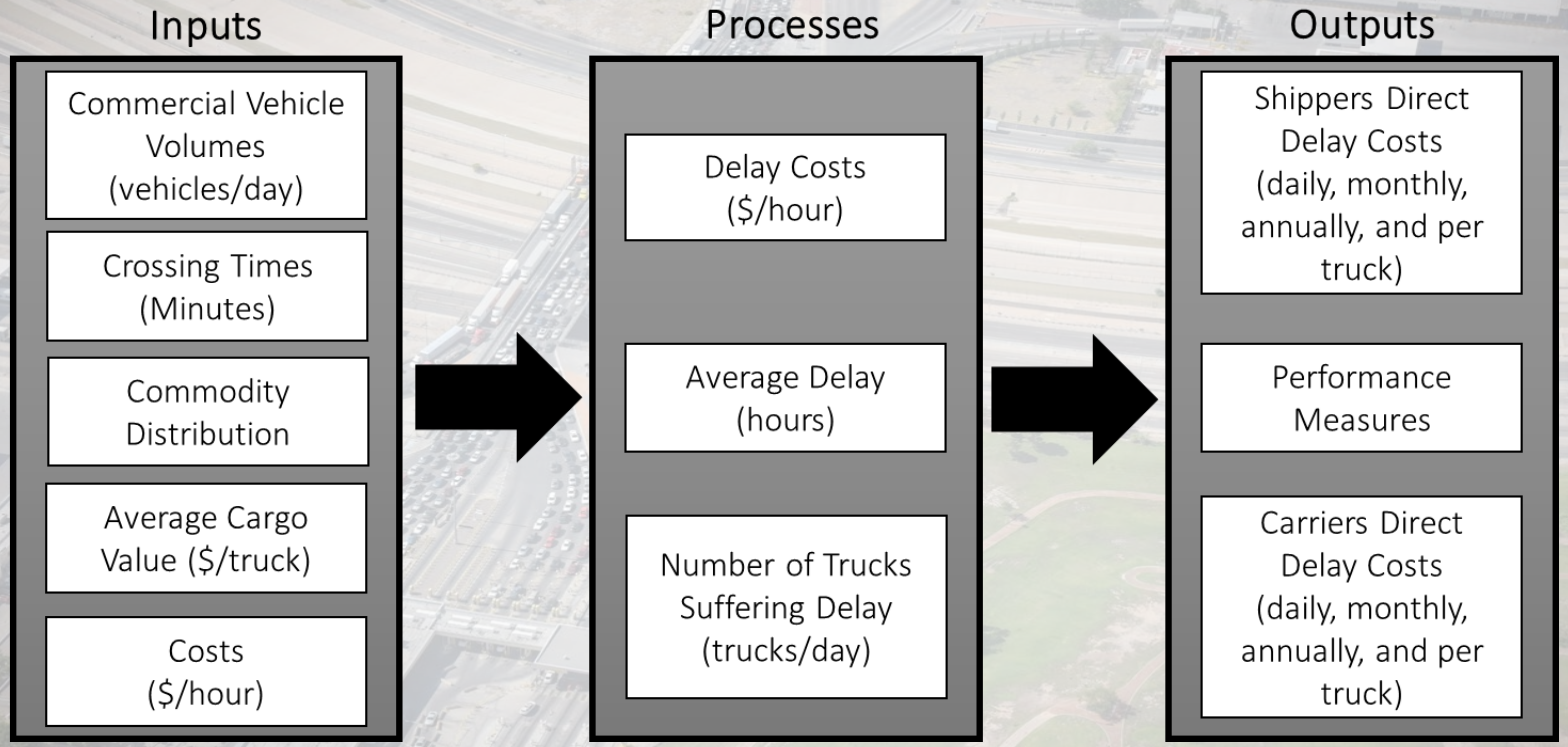
# Background: Motivation

- **The direct cost of border delays varies based on the commodities carried across each LPOE, which vary across Texas LPOEs**
  - Just-In-Time (JIT) commodities have higher cost of delay than non-JIT
  - The El Paso region LPOEs carry a significant amount of JIT cargo
  - The DCET allows incorporating these costs into estimates
  - El Paso MPO is interested in leveraging the work on SB 1907 by expanding the scope of analysis to compare cost of total delay across LPOEs in the El Paso, Laredo and the Rio Grande Valley

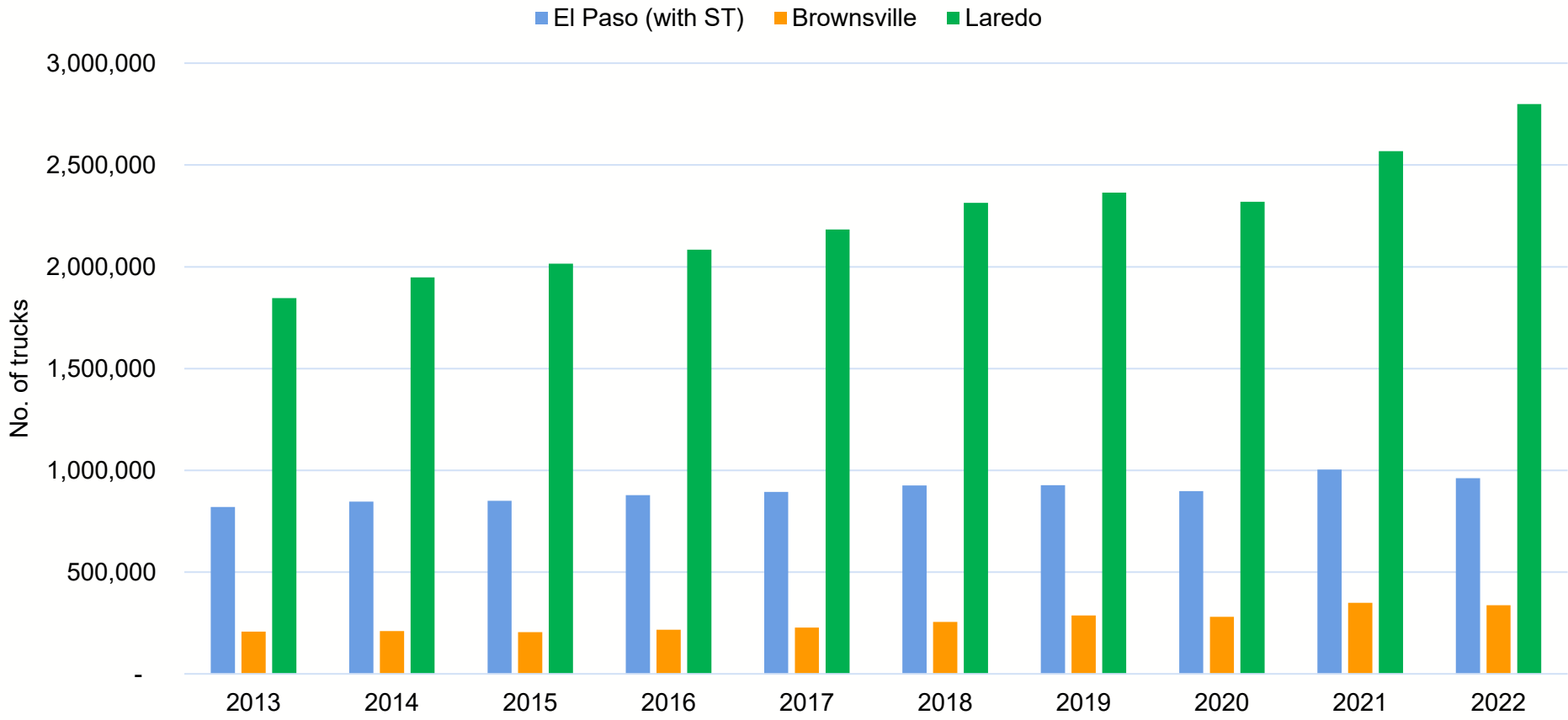
# Objective

- **Compare the impacts of border crossing delays across five Texas LPOEs by incorporating the value and volume of the commodities being transported into the cost of delay calculation.**
  - Demonstrate the influence that cargo mix in each LPOE has on the costs of delay
  - Provide a more comprehensive framework to evaluate and prioritize improvements aimed at reducing the costs of delays
- **LPOEs include:**
  - Bridge of the Americas (BOTA) in El Paso;
  - Ysleta-Zaragoza Bridge in El Paso;
  - World Trade Bridge in Laredo;
  - Laredo-Colombia Solidarity Bridge in Laredo;
  - Veterans-Los Tomates Bridge in Brownsville;
  - Santa Teresa Bridge in Santa Teresa.

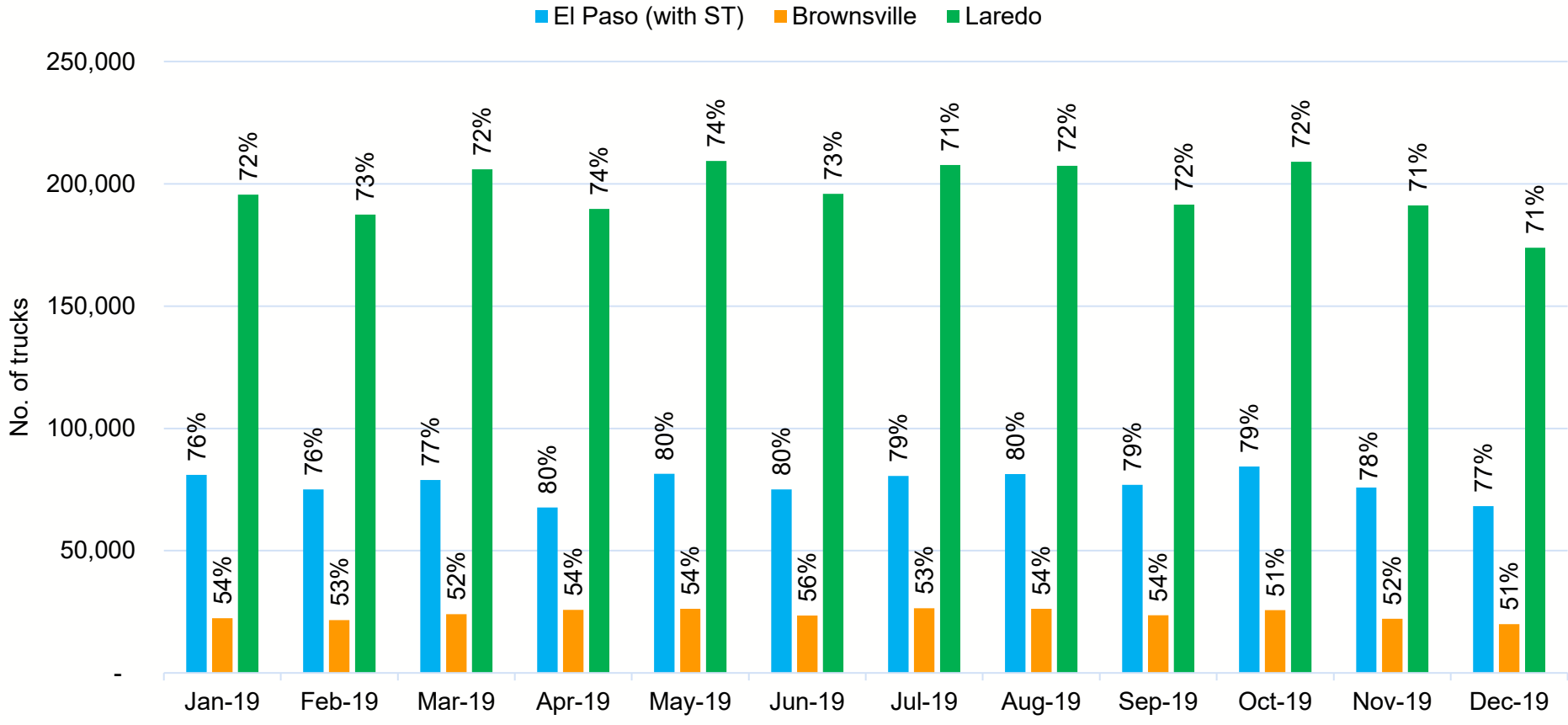
# Direct Cost Estimation Tool



# Commercial Vehicle Volumes (No. of trucks)

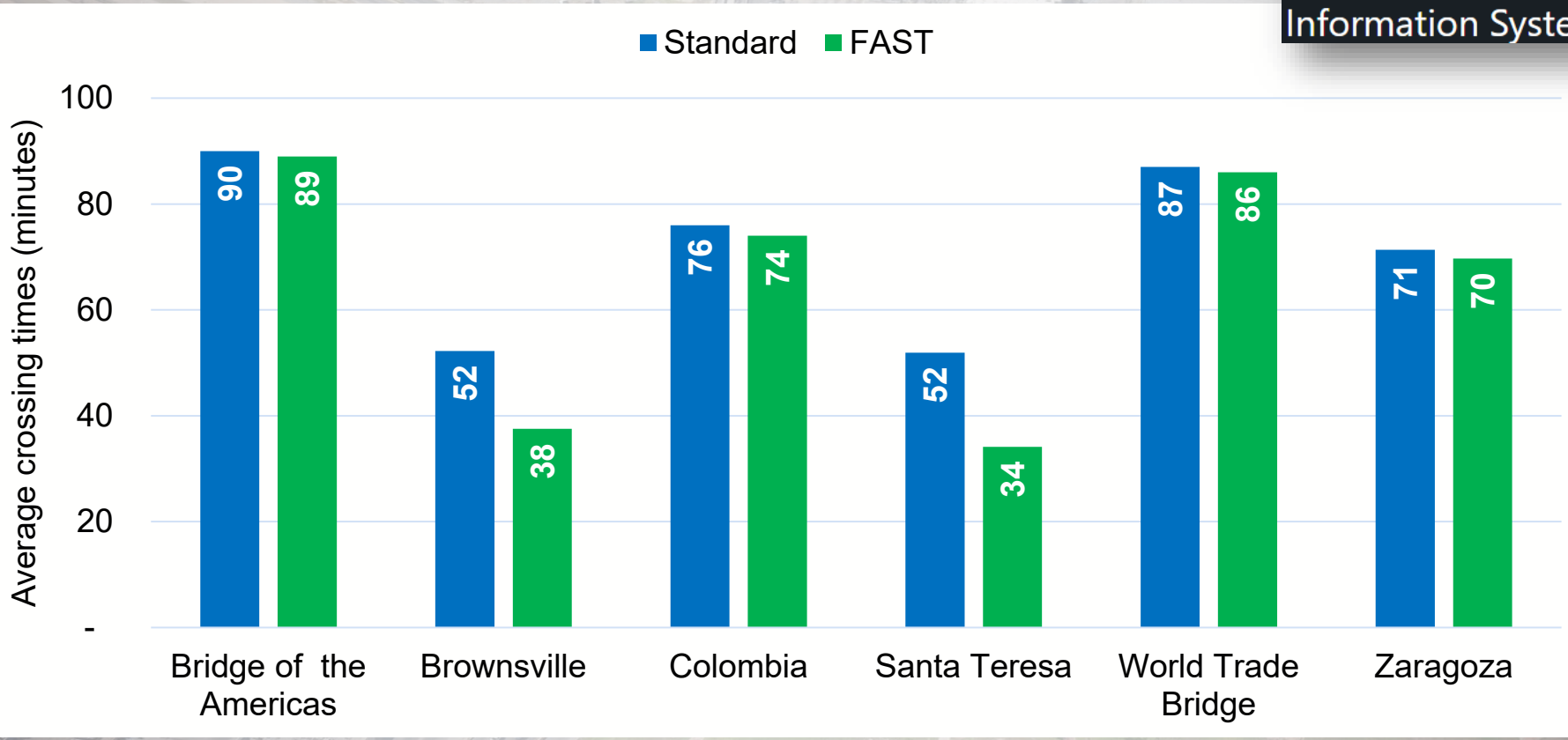


# Commercial Vehicle Volumes (with loaded truck rates)



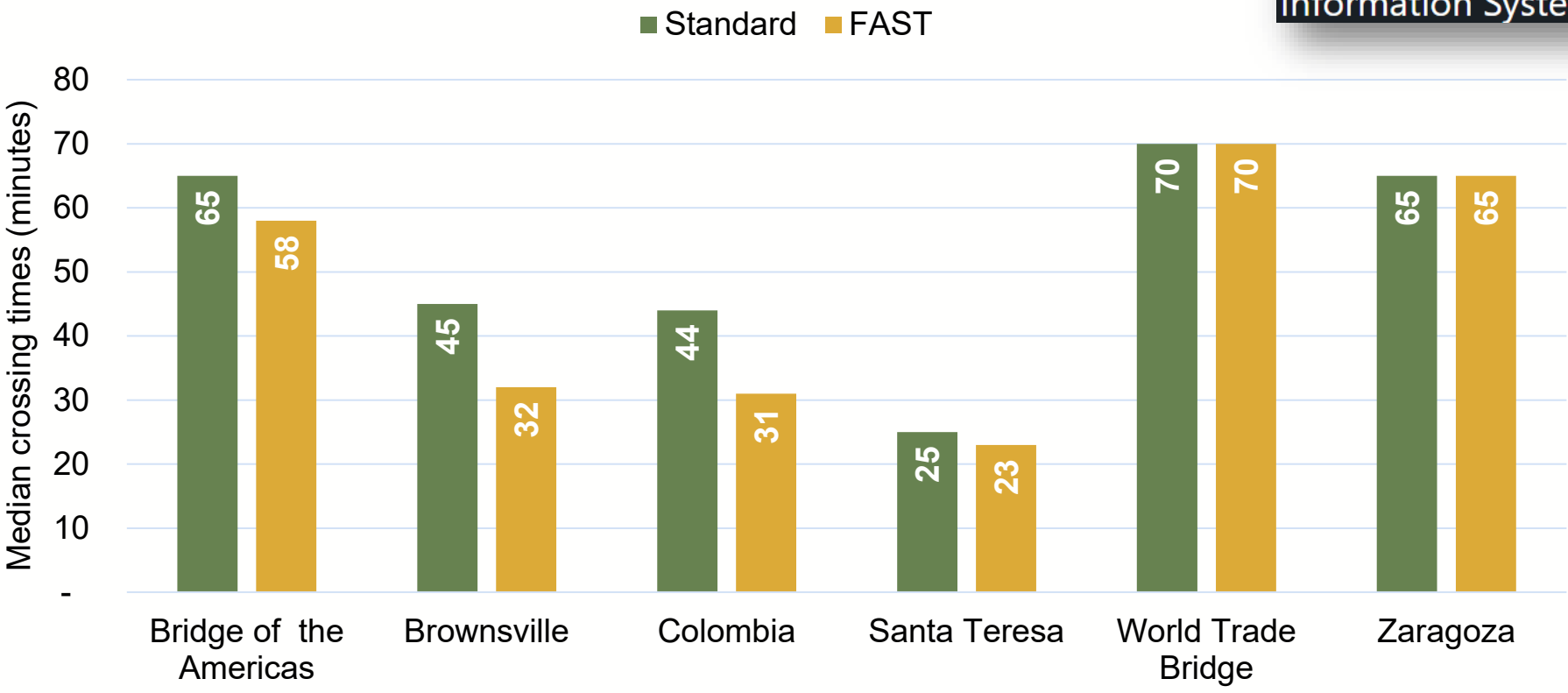
# Average Crossing Times

Border Crossing Information System



# Median Crossing Times

Border Crossing Information System



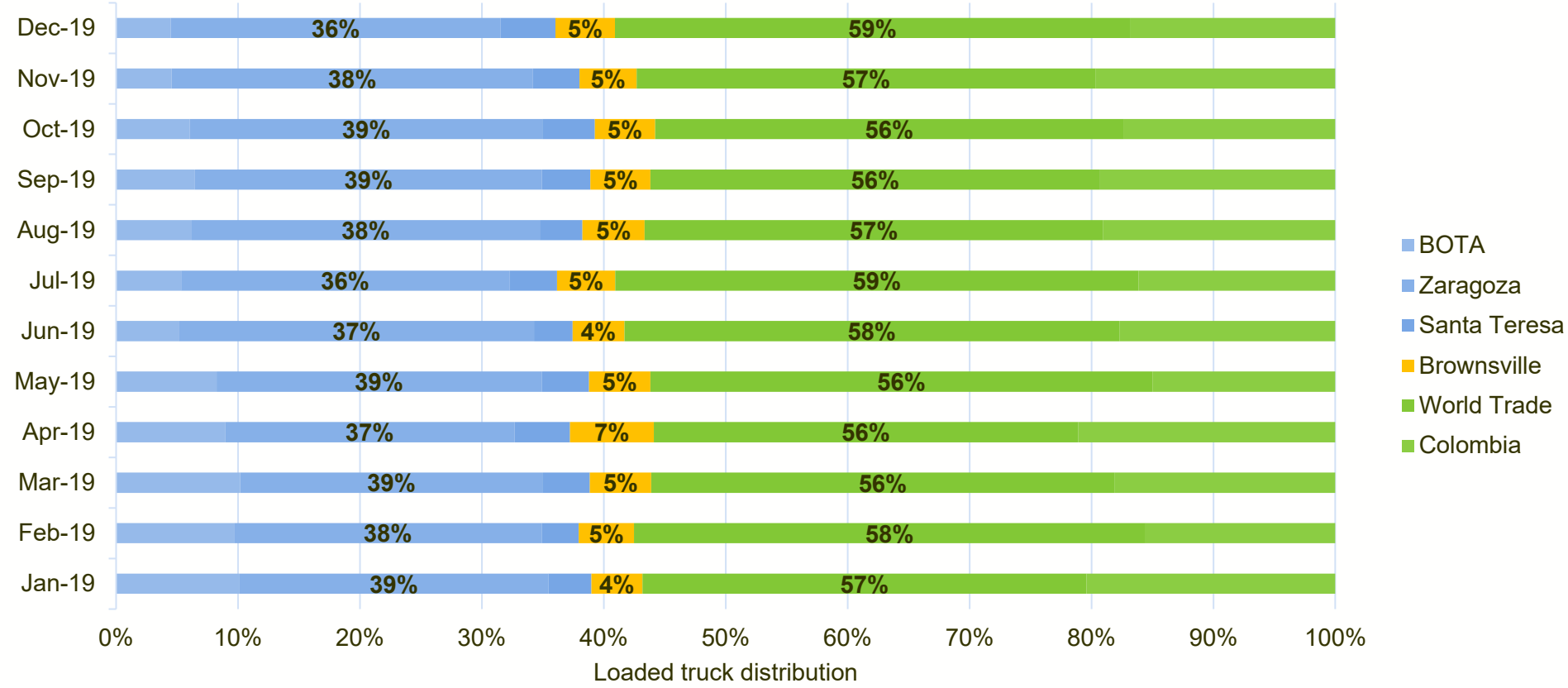
# Commodity Distribution



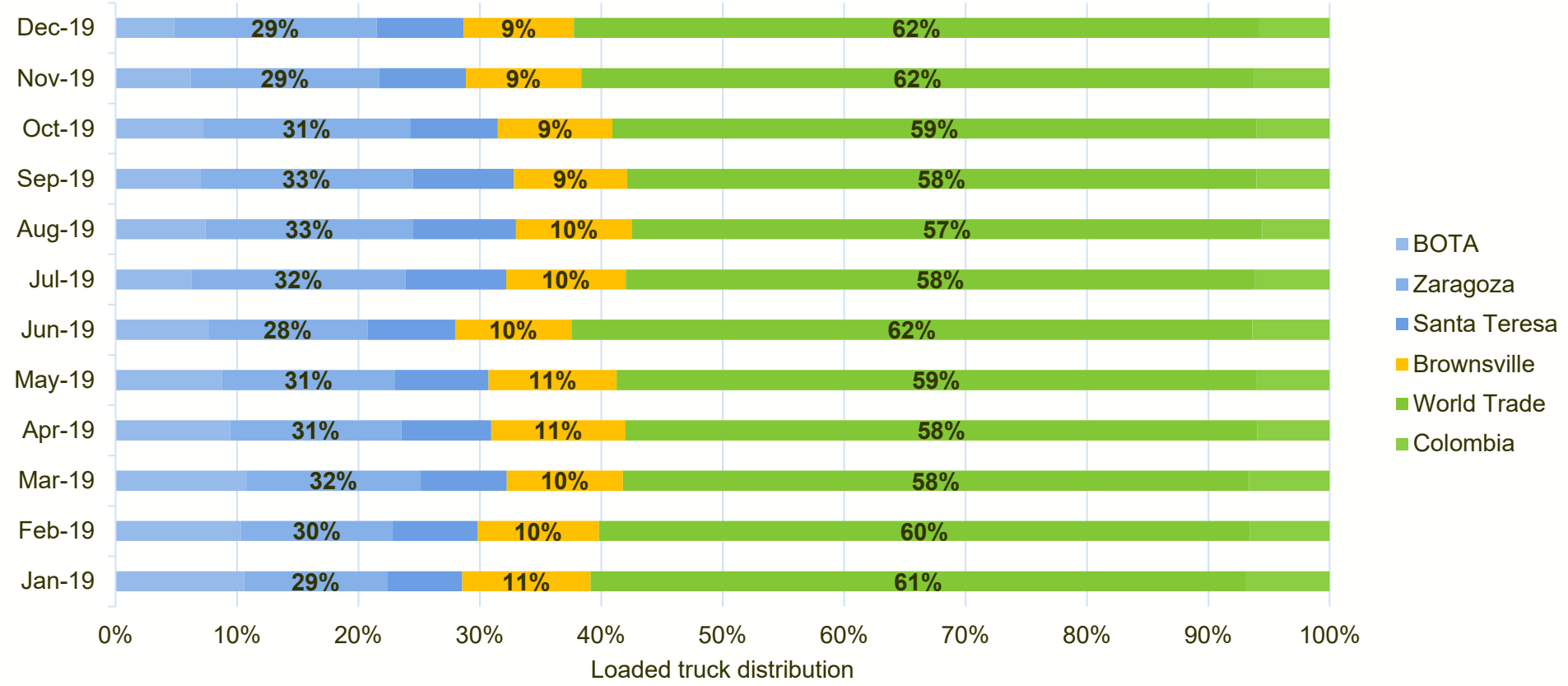

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**Bureau of Transportation Statistics**

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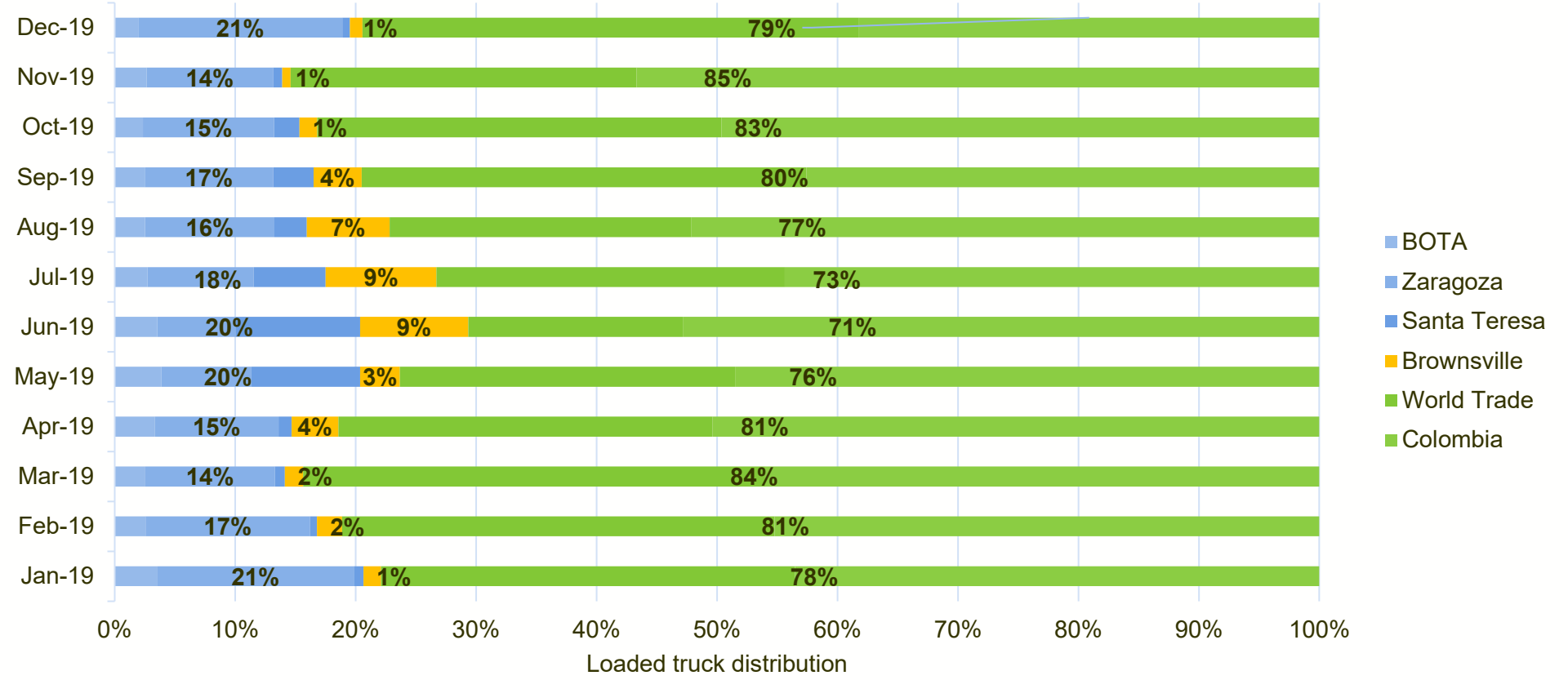
# Commodity Distribution (just-in-time)



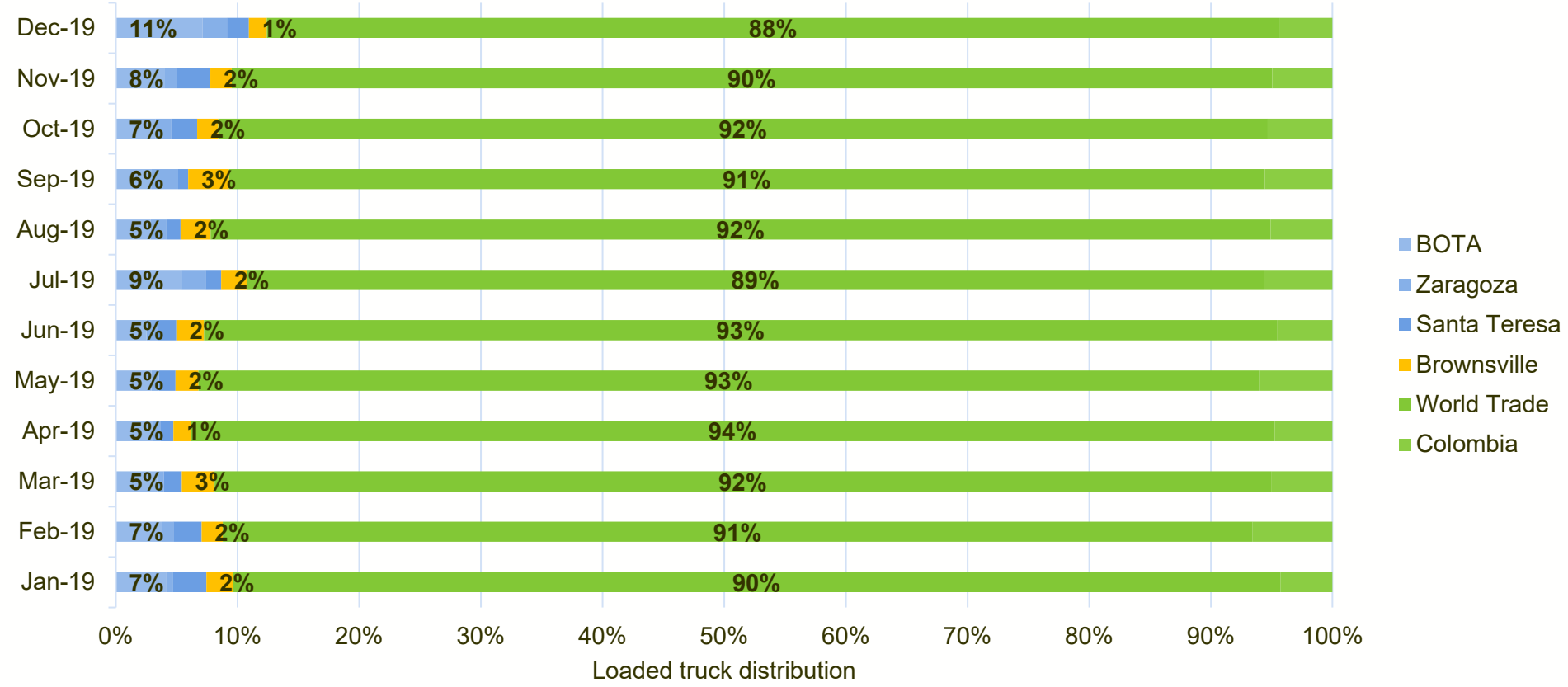
# Commodity Distribution (non-just-in-time)



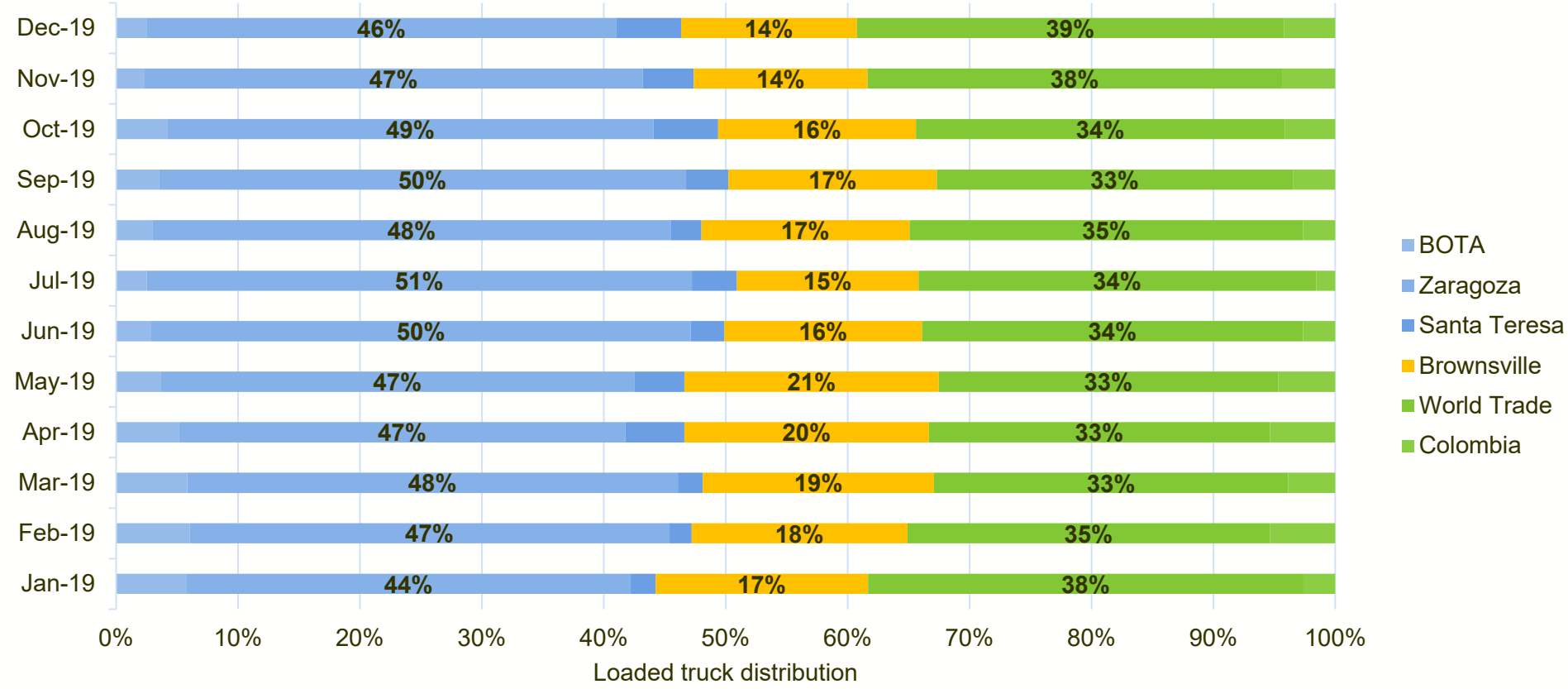
# Commodity Distribution (perishables)



# Commodity Distribution (non-perishables)

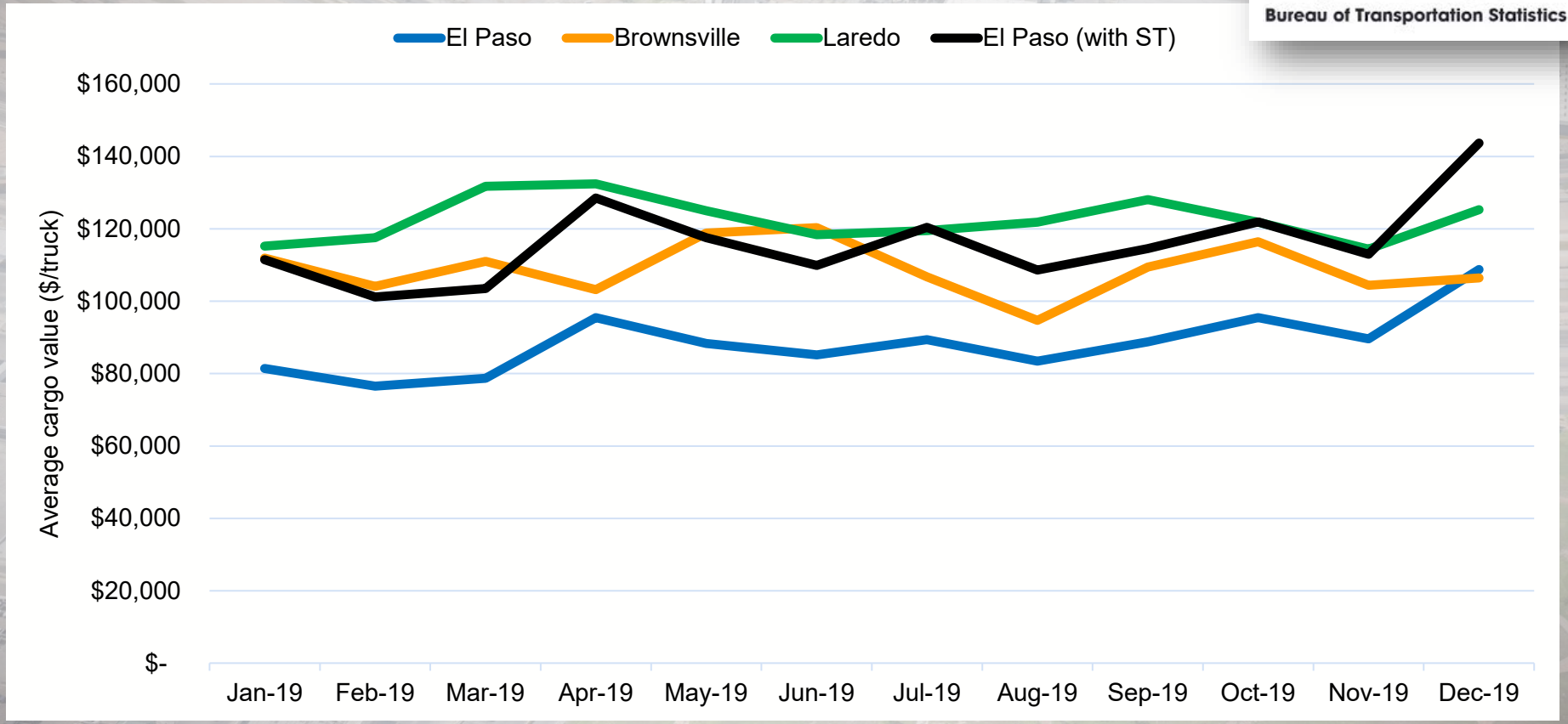


# Commodity Distribution (other)



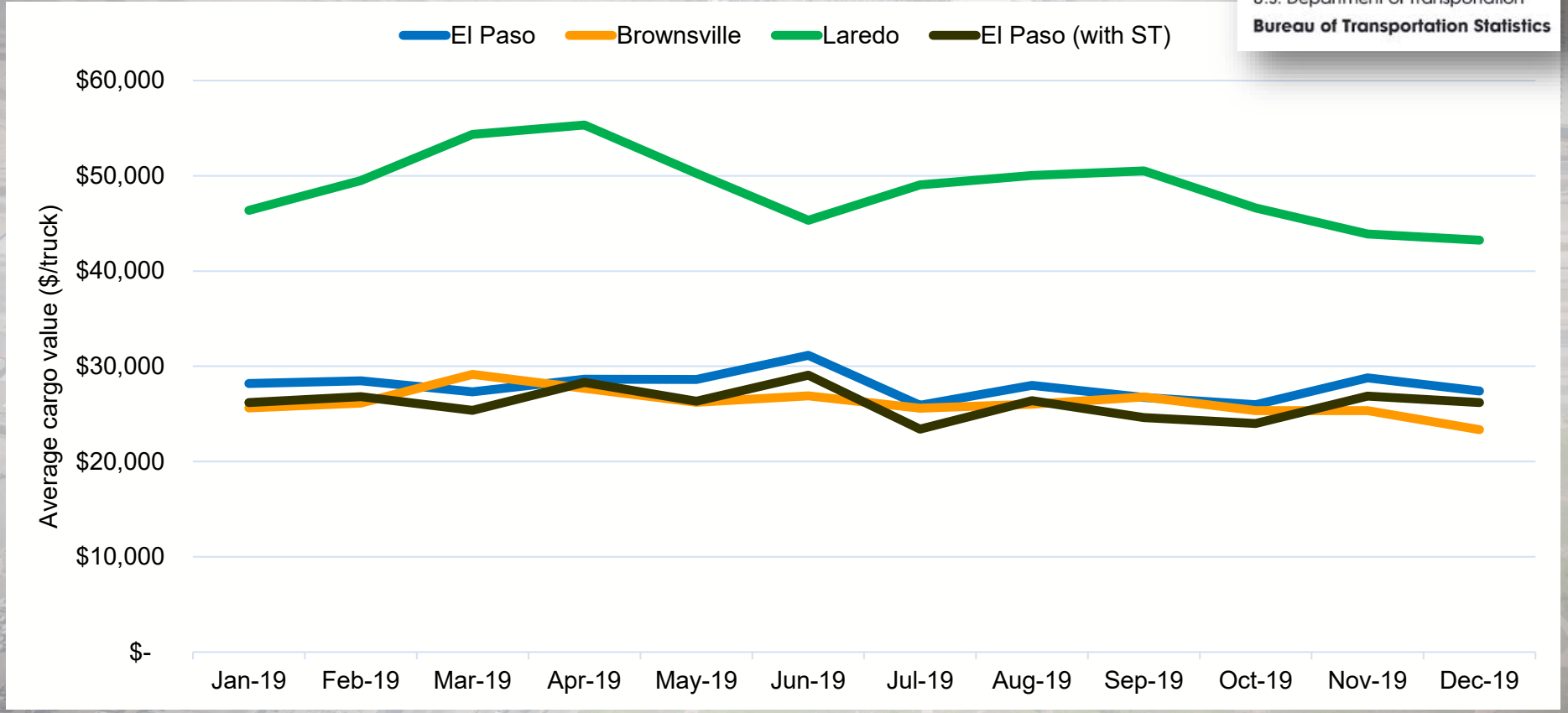
# Average Cargo Value (just-in-time)

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**Bureau of Transportation Statistics**



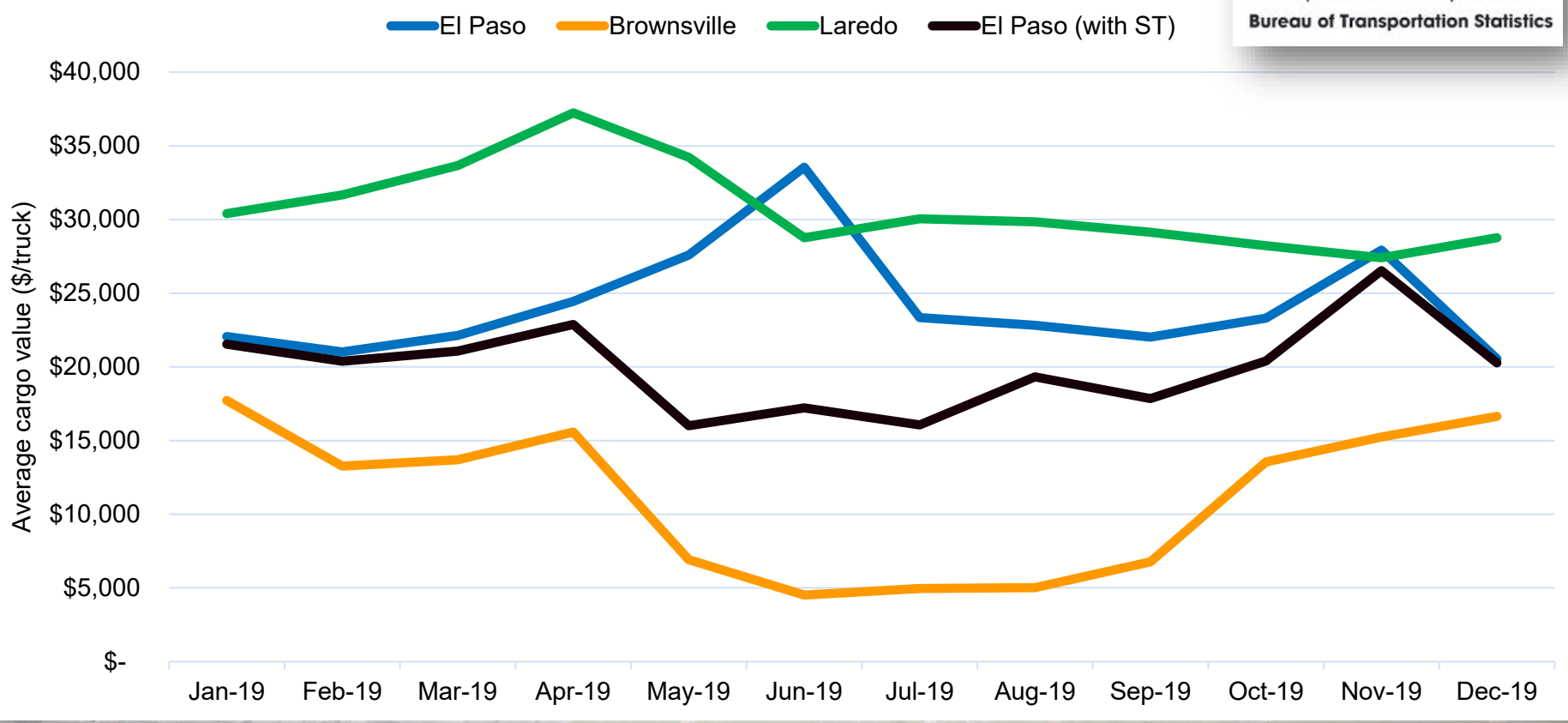
# Average Cargo Value (non-just-in-time)

U.S. Department of Transportation
**Bureau of Transportation Statistics**



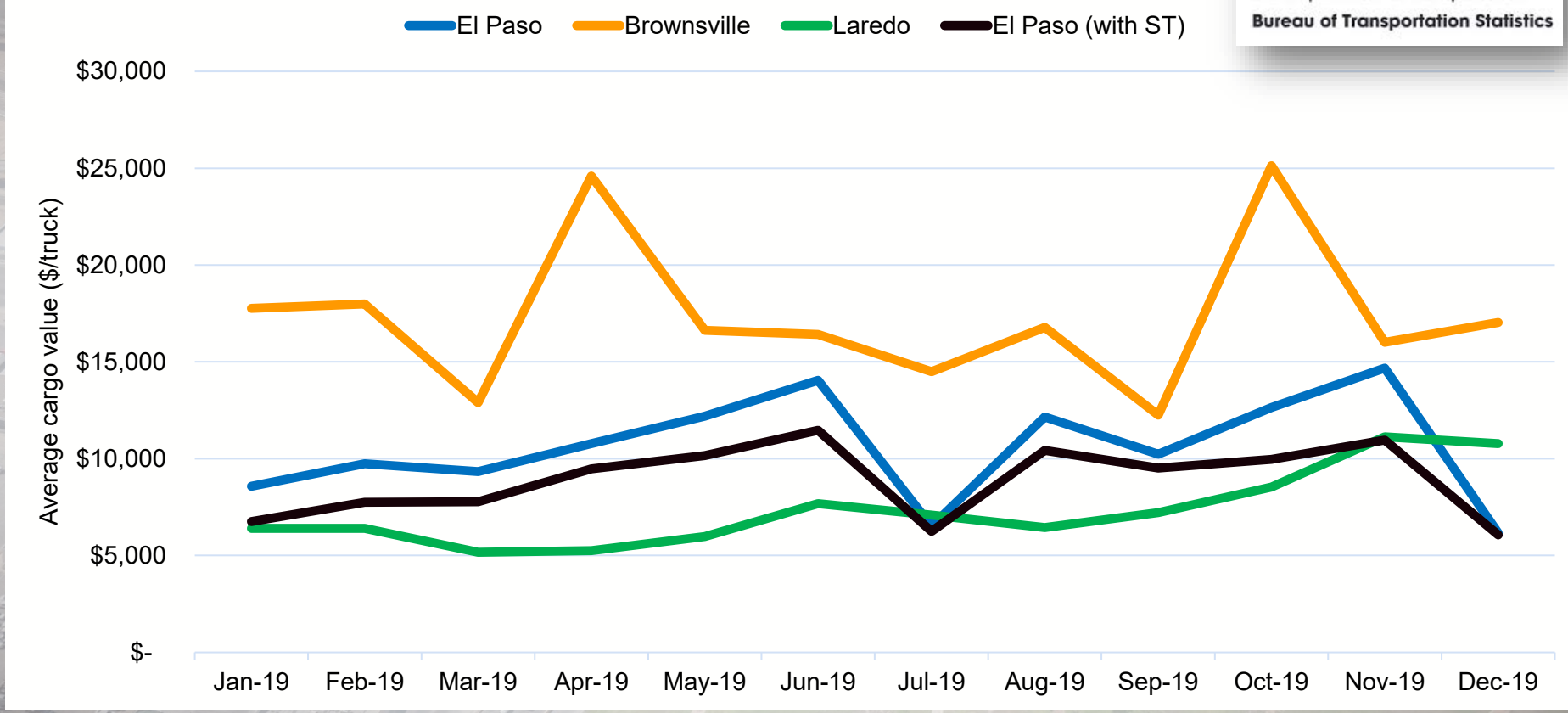
# Average Cargo Value (perishables)

U.S. Department of Transportation
**Bureau of Transportation Statistics**



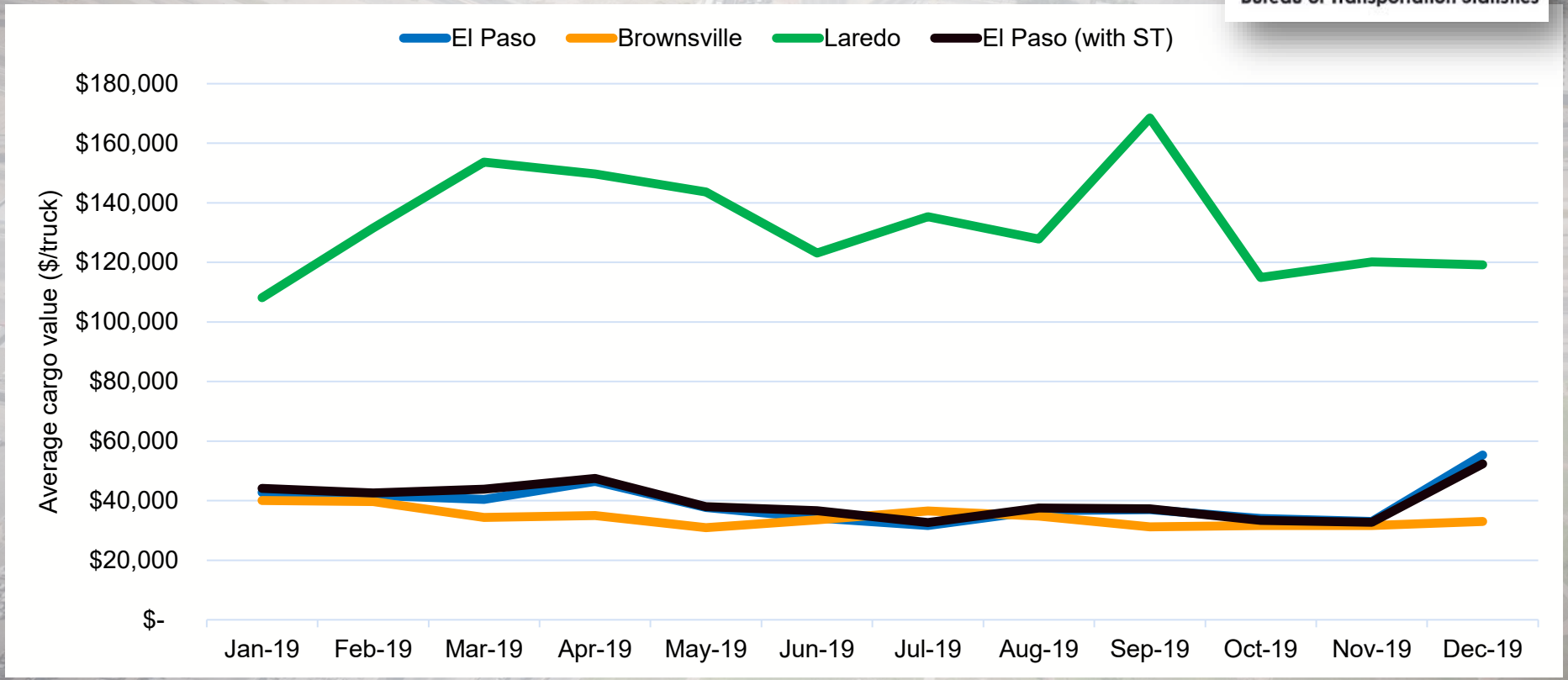
# Average Cargo Value (non-perishables)

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**Bureau of Transportation Statistics**



# Average Cargo Value (other)

U.S. Department of Transportation
**Bureau of Transportation Statistics**



# Costs (2019)

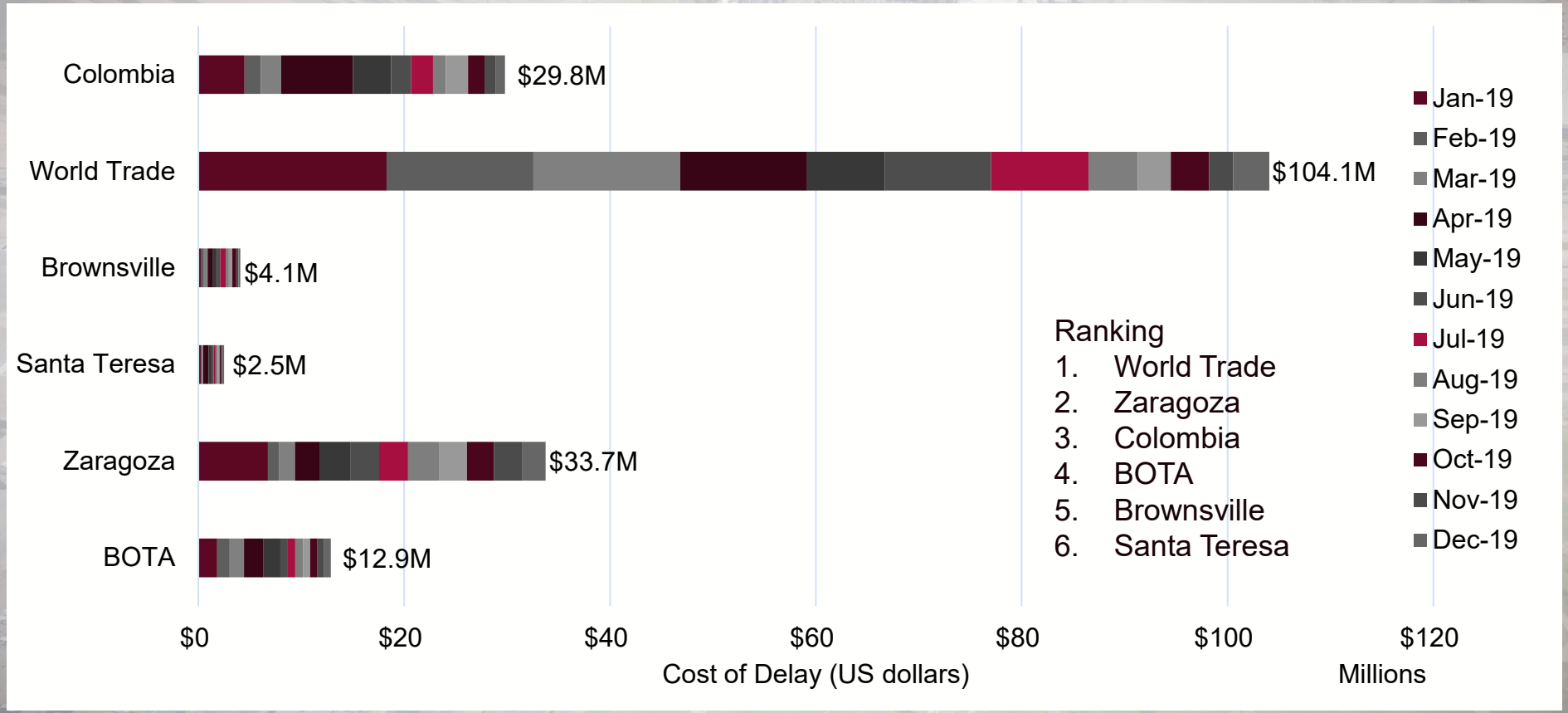
## Shipper costs

| | |
|---|---|
| Inventory Cost | • Inventory Risk Cost:18 % of average cargo value<br>• Capital Cost Interest:3.25% of average cargo value |
| Additional Logistics Cost | • Reliability Cost for Buffer Time: $168.53/hour<br>• Reliability Cost for Schedule Delay: $371.33/hour |

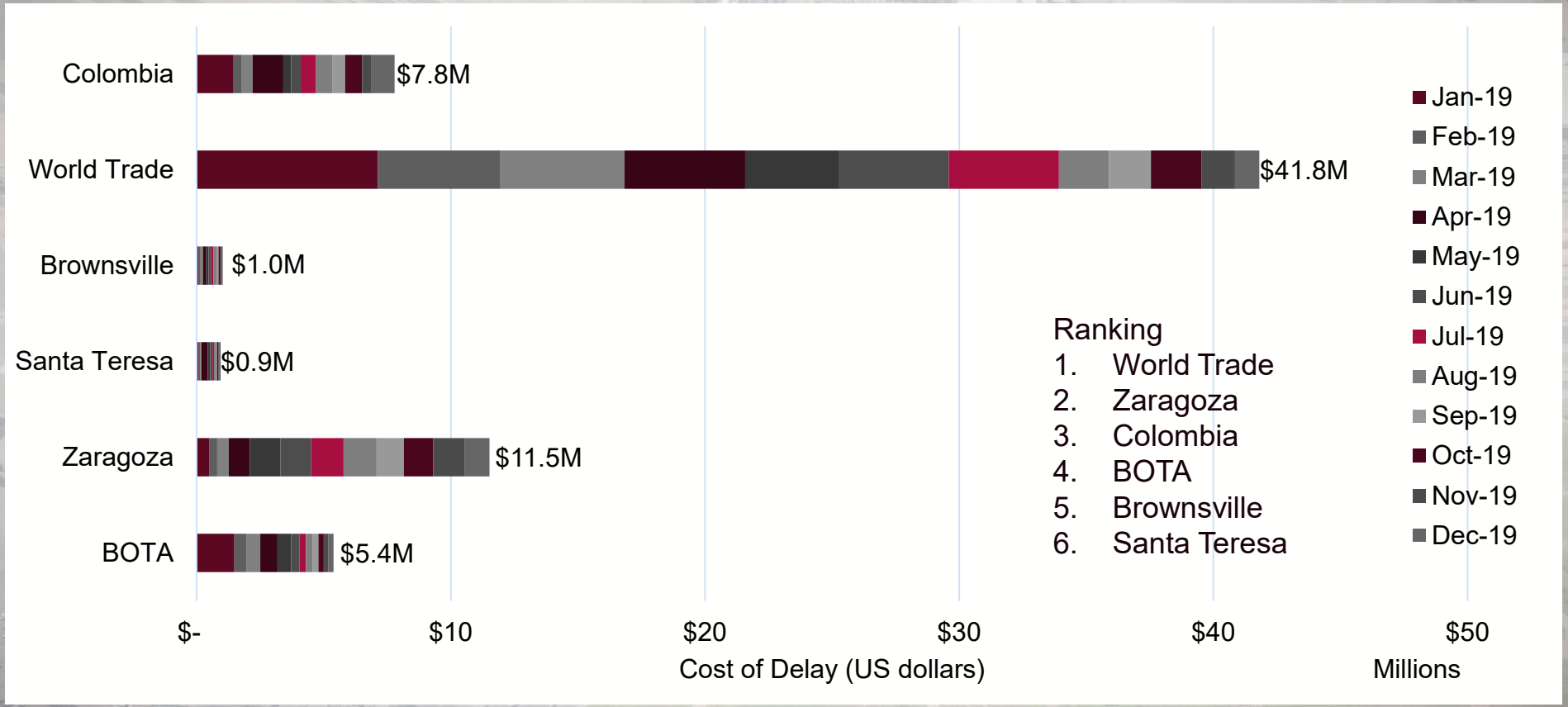
## Carrier costs

| | |
|---|---|
| Fuel price | • Updated monthly |
| Driver wages | • El Paso<br>• Laredo<br>• Brownsville |

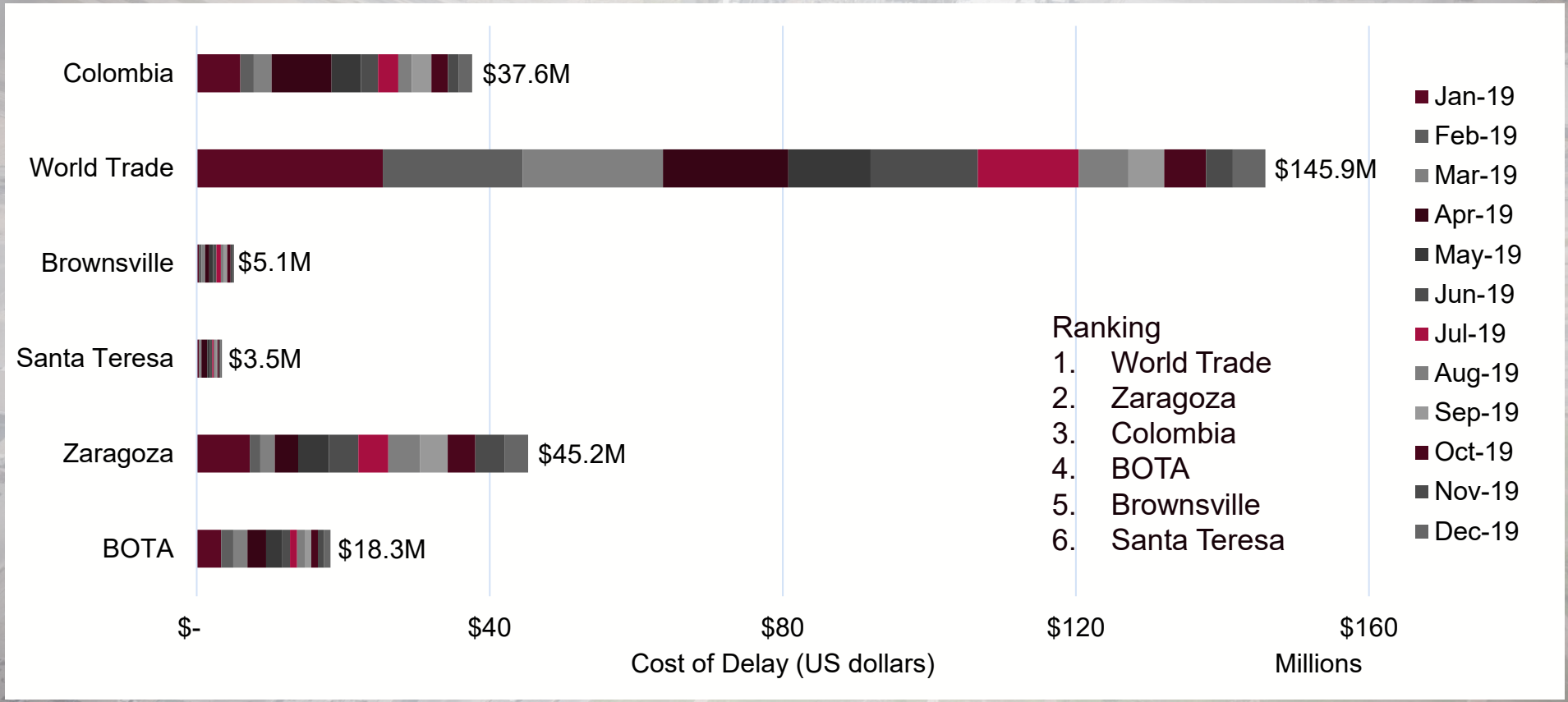
# DCET Outputs (Standard)



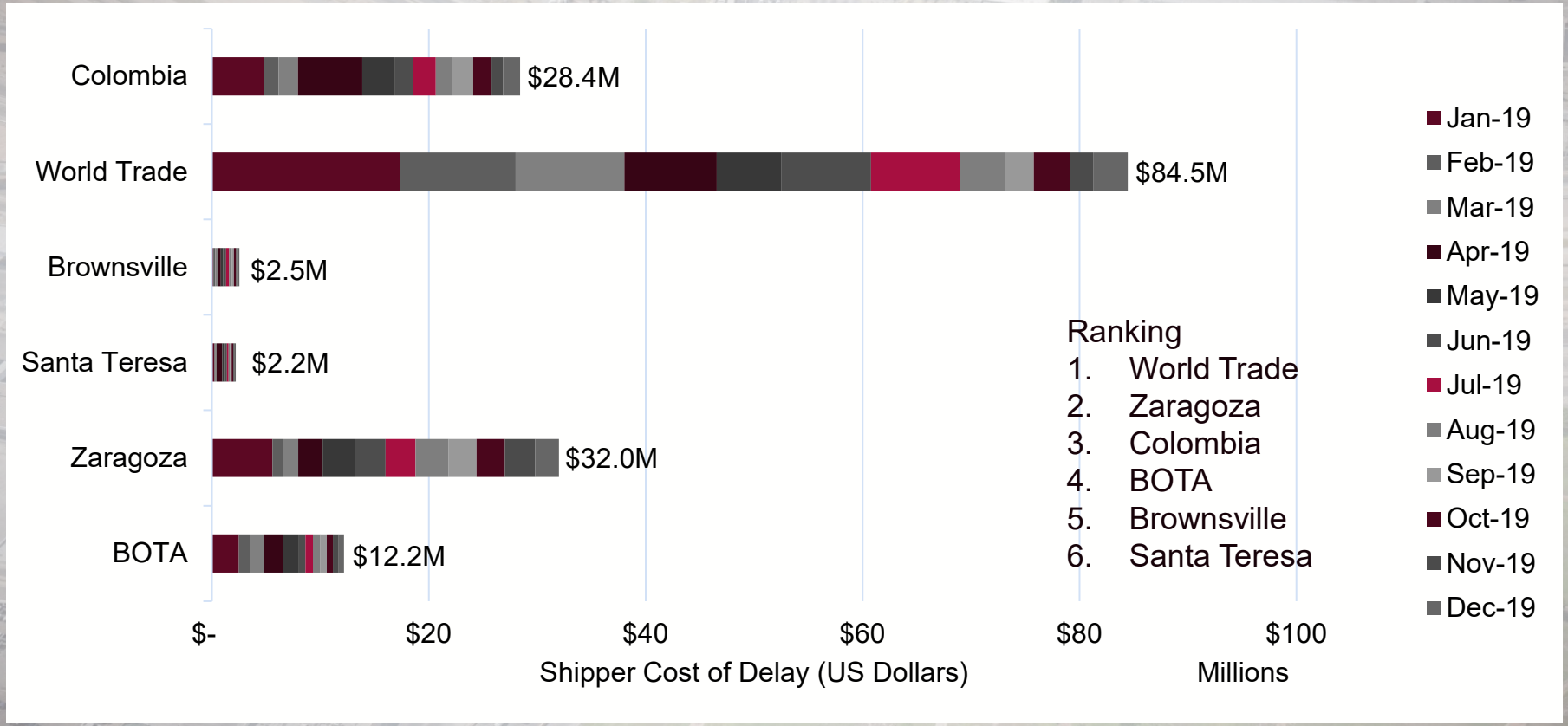
# DCET Outputs (FAST)



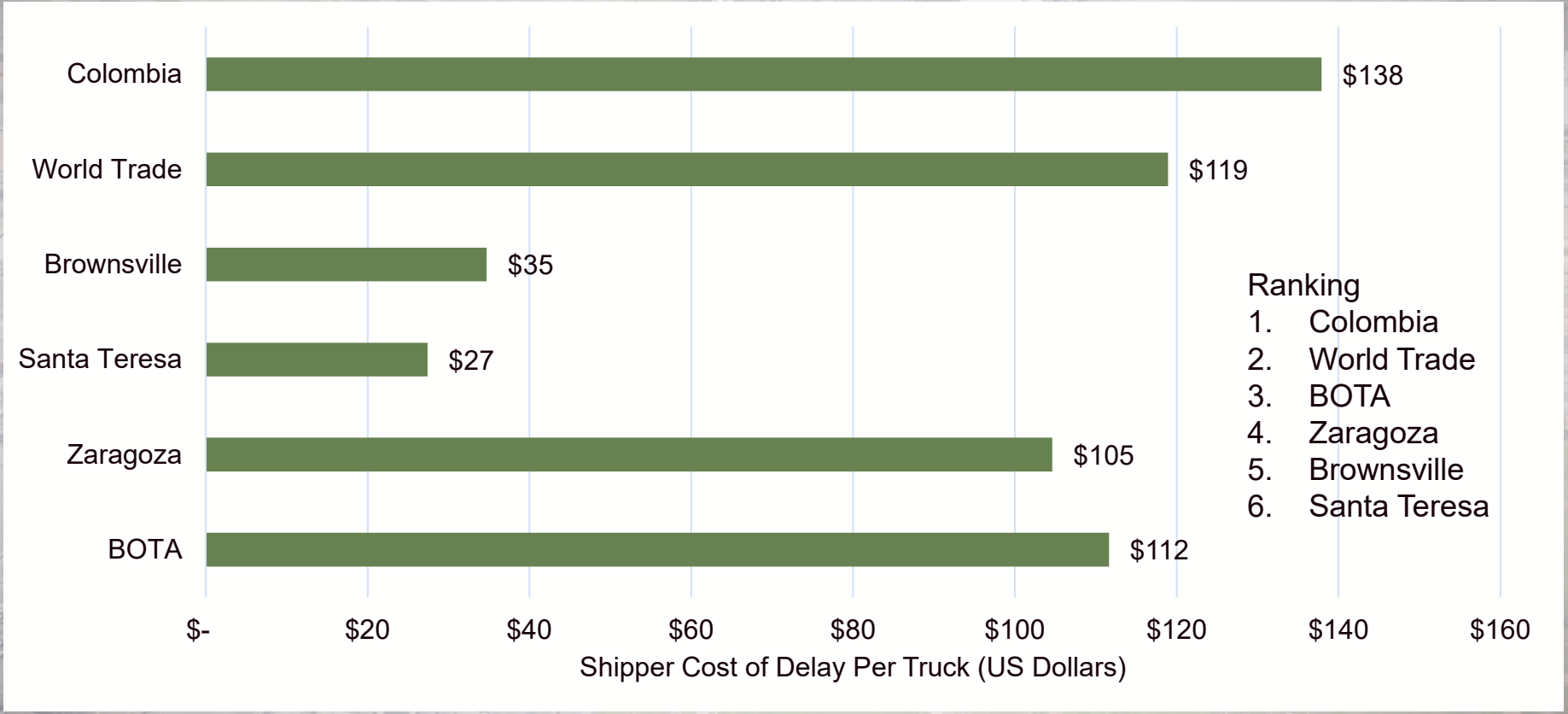
# DCET Outputs (Total)



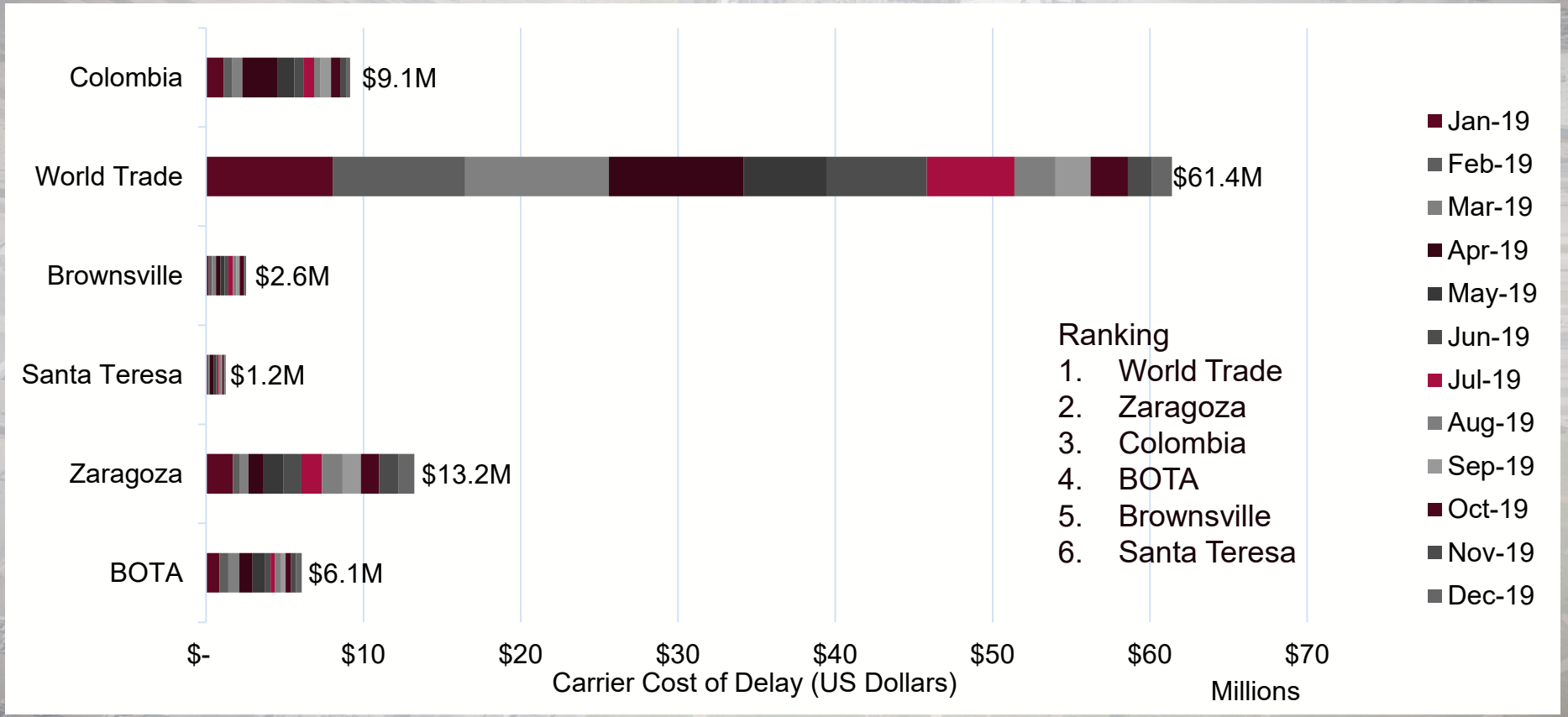
# DCET Outputs (Shipper Costs-Total)



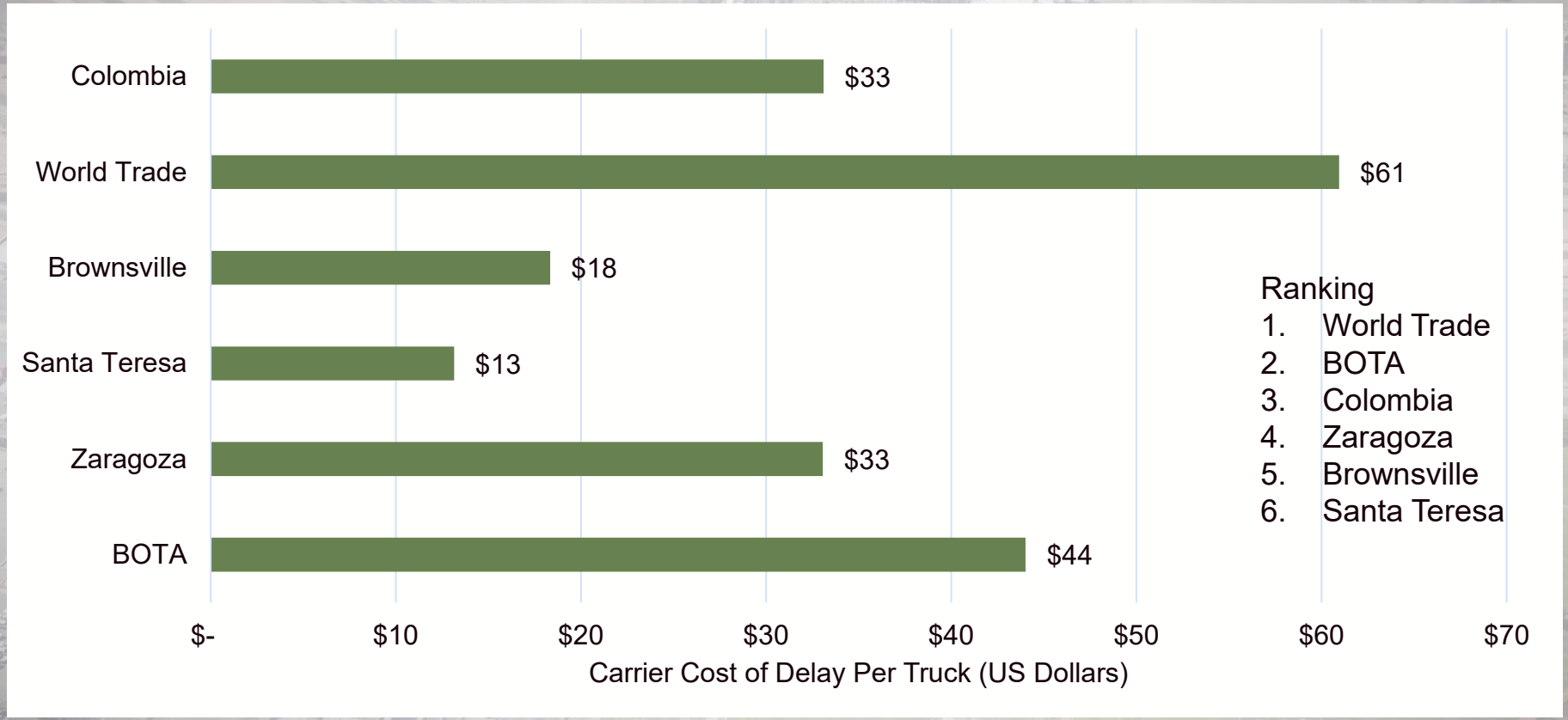
# DCET Outputs (Shipper Costs-Average Cost per Truck)



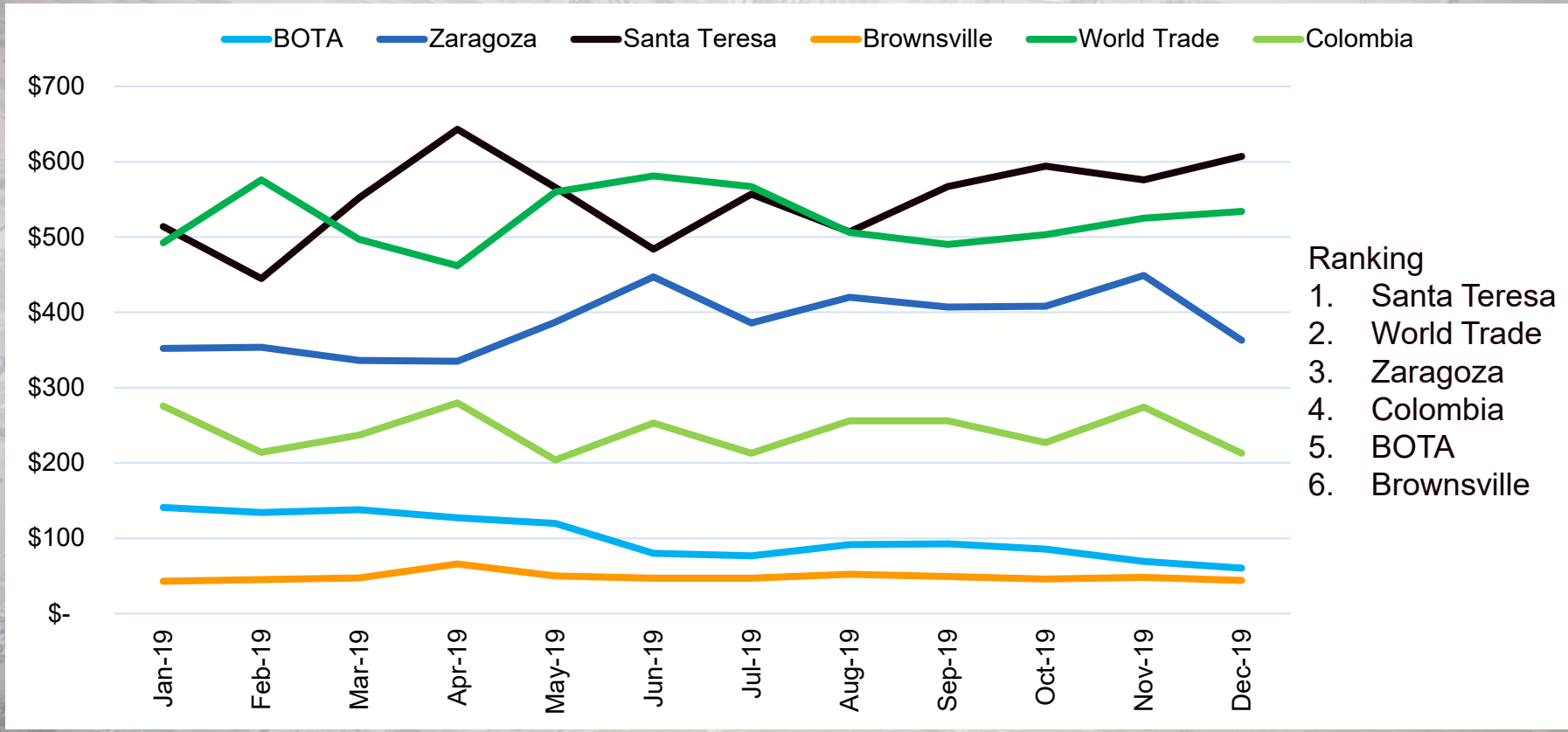
# DCET Outputs (Carrier Costs-Total)



# DCET Outputs (Carrier Costs-Average Cost per Truck)



# DCET Outputs (Unit Average Delay Cost)
# 1 minutes of delay for 1 truck




Ranking

1. Santa Teresa
2. World Trade
3. Zaragoza
4. Colombia
5. BOTA
6. Brownsville

# THANK YOU




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